

Memory and Long-Context Mechanisms for Long-Horizon LLM Agents

Abstract

Long-horizon large language model agents must operate over interaction histories, retrieved evidence, tool traces, environmental observations, and self-generated reasoning that quickly exceed the practical limits of a single prompt. This makes memory neither a peripheral enhancement nor merely a user-experience feature, but a central systems problem that couples representation, retrieval, compression, training, inference efficiency, and evaluation. Recent work therefore spans two partially overlapping traditions: long-context language modeling, which attempts to enlarge or restructure the model’s effective working context, and agent memory, which attempts to externalize, consolidate, and selectively reuse information across episodes, sessions, and tasks [7–17]. ¹

This survey organizes the area around a simple claim: **long-horizon agents need both more bandwidth and better persistence**. Architectural long-context mechanisms increase how much evidence can be read within one forward pass, while memory systems determine what should survive beyond that pass, in which form, and under which retrieval policy. Read through this lens, ostensibly disparate lines of work—recurrent transformers, sparse attention, state-space models, retrieval-augmented language models, conversational memory banks, reflective memory management, profile and skill memory, prompt compression, and KV-cache optimization—become instances of a shared design space balancing fidelity, latency, cost, updateability, and robustness [8–13, 18–69, 78–97]. ²

The survey argues that the field is moving from **storage** to **experience**. Early systems largely preserved raw trajectories or extended context windows. More recent systems attempt to transform trajectories into semantically indexed, hierarchically organized, or procedurally reusable memory objects—facts, summaries, notes, plans, skills, reflections, or graph structures—that can support continual interaction and adaptation. Yet current evidence also shows that strong needle-style retrieval does not imply reliable long-context reasoning, and large context windows do not by themselves solve selective remembering, knowledge updating, or multi-session coherence. These gaps define the central research agenda for long-horizon LLM agents [12, 40–50, 64–69, 78–97, 98–132]. ³

Introduction

The core difficulty in long-horizon agency is that an agent’s relevant state is distributed across time. A coding agent may need to remember earlier compilation failures and repository conventions; a web agent may need to preserve facts about page dynamics and hidden interface affordances; a personalized dialogue agent may need to track preferences, goals, exceptions, and life events across sessions; an embodied agent may need to accumulate reusable skills rather than re-derive them from scratch. In all such cases, the agent must simultaneously maintain a **working memory** for the current control loop and a **persistent memory** for information worth carrying forward. The historical roots of this problem appear in recurrent modeling, external neural memory, and memory-augmented sequence models [1–6], but the modern agentic formulation only became explicit after the rise of transformer LLMs and prompt-based reasoning systems [7–9, 12–17]. ⁴

A useful distinction is between **long context** and **memory**. Long context asks how much information can be processed within a single inference episode, typically by scaling attention, altering positional representations, or restructuring sequence processing. Memory asks what information should remain available after the episode, how it should be abstracted, and how it should later be retrieved or updated. The two are complementary rather than interchangeable. Long-context models can read more, but they still suffer from position sensitivity and inference cost; memory systems can persist information efficiently, but they can introduce retrieval errors, stale facts, and maintenance burdens [12–13, 33–50, 91–97]. Lost-in-the-middle results, conversational-memory benchmarks, and recent cost-performance analyses all suggest that simply increasing prompt length does not eliminate the need for explicit memory management [12, 90–97]. ⁵

This survey therefore treats memory for long-horizon agents as a layered problem. At the lowest level are architectural mechanisms that determine the effective size and structure of the active context. On top of that are non-parametric memory stores and retrieval systems that provide access to externalized knowledge. Above these sit agent-level memory organizations—episodic, semantic, procedural, reflective, or personalized—that decide what to write, when to consolidate, how to forget, and how to reuse past experience. Finally, evaluation determines which combinations of these mechanisms actually improve downstream agent competence rather than merely benchmark-local retrieval [13–17, 41, 64–69, 78–109]. To keep the discussion focused, the survey prioritizes papers directly relevant to these mechanisms rather than broad adjacent literatures on LLM prompting or generic information retrieval.

⁶

Methods

Architectural long-context extension is the first family of approaches. Classical transformer scaling confronts the quadratic cost of full attention, which motivated sparse or approximated attention mechanisms such as Sparse Transformers, Reformer, Longformer, BigBird, Linformer, and Performers [18–24]. Another line revisits recurrence or compressed hidden state, from Transformer-XL and Compressive Transformer [8–9] to later recurrent or state-based alternatives including S4, Hyena, RetNet, Mamba, and hybrid architectures such as Jamba [26–30]. A third line targets positional extrapolation and context extension directly through ALiBi, RoPE-based interpolation, YaRN, LongLoRA, LongQLoRA, Self-Extend, LongRoPE, and LongRoPE2 [10–11, 33–39]. These methods differ in where they pay the cost: some reduce asymptotic complexity, some preserve full attention only locally, and some keep the architecture fixed while changing how positions or adaptation data are used. Parallel context encoders, landmark-based random access, and streaming approaches such as CEPE, Landmark Attention, and StreamingLLM illustrate an increasingly pragmatic trend: rather than insisting on uniform full attention everywhere, they introduce structured side channels for accessing long-range information [42, 45, 47].

⁷

For long-horizon agents, however, bigger windows are useful only insofar as they preserve task-relevant access patterns. This is where efficient prefill and KV-centric methods matter. FlashAttention and FlashAttention-2 reduce the systems cost of dense attention [31–32], while H2O, SnapKV, and MInference selectively retain or sparsify attention states to improve long-context inference throughput [48–50]. Activation Beacon, gist-style compression, and related latent-compression methods move one step further by learning or inducing compressed intermediate representations that can be treated as portable short-term memory objects [127–128]. The important conceptual point is that these methods blur the line between “context processing” and “memory”: once activations, beacons, gist vectors, or retained KV segments become reusable across turns, the active context itself becomes a memory substrate. Yet benchmark evidence also shows that aggressive query-conditioned compression can

degrade shared-context or multi-turn performance, because what looks irrelevant to the current query may be needed later [109]. ⁸

External non-parametric memory and retrieval forms the second method family. The central idea is to keep the backbone model relatively fixed while giving it access to an external datastore keyed by embeddings, internal activations, summaries, or symbolic structures. kNN-LM, REALM, RAG, FiD, Atlas, and RETRO showed that external retrieval can improve factual grounding, adaptation, and long-tail behavior without requiring all relevant knowledge to be stored parametrically [51–56]. Memorizing Transformers, Memformer, LongMem, MEMORYLLM, M+, MemLong, and Extended Mind Transformers extend this principle toward sequence modeling itself: retrieval is no longer only about external documents, but about re-accessing previous hidden states, cached chunks, or trainable latent memory pools [57–63]. In agents, this family is attractive because it supports persistence, updateability, and lower amortized cost than repeatedly replaying full histories. But it also creates a familiar systems problem: indexing quality, memory granularity, and query formulation become part of the agent’s reasoning pipeline rather than mere preprocessing. ⁹

Recent retrieval systems increasingly organize memory hierarchically or relationally rather than as a flat vector store. REPLUG and Self-RAG refine the interaction between retriever and generator [64–65], while RAPTOR, HippoRAG, MemoRAG, and LightRAG use recursive summarization, graph structure, or global memory cues to improve multi-hop access over long corpora [66–69]. For long-horizon agents, these designs are especially important because future queries are often underspecified. A later question may depend on an inferred goal, a latent preference, or a previously failed strategy rather than a lexically matching fact. Graph- and hierarchy-aware memory therefore functions not just as storage, but as **abductive recall**: it supports reconstruction of what mattered, not only exact replay of what was said or seen. This is one reason recent agent-memory papers increasingly borrow conceptual vocabulary from episodic and semantic memory rather than treating conversation history as a plain document retrieval problem [16–17, 67–69, 82–89]. ¹⁰

Agent memory stacks and experience abstraction constitute the third family. Generative Agents provided an influential early template: maintain a stream of experiences, retrieve by relevance and recency, generate higher-level reflections, and use these memories for planning [70]. Reflexion and ExpeL showed that language feedback and distilled experiences can themselves become memory objects that improve later behavior without parameter updates [71–72]. Voyager extended this idea to embodied lifelong learning by storing executable skills in a growing library, making procedural memory a first-class component of open-ended adaptation [73]. ReAct, Tree of Thoughts, RAP, and LATS are usually framed as reasoning or planning methods [74–77], but for long-horizon agents they also define what intermediate state is worth externalizing: deliberation traces, search trees, failure analyses, solver state, and reusable subplans. Under this reading, planning and memory are tightly coupled. Planning determines which abstractions are useful; memory determines whether those abstractions survive beyond one attempt.

¹¹

In personalized or conversational settings, agent memory becomes more explicitly structured. MemoryBank and MemGPT introduced long-term memory mechanisms that maintain evolving records across interactions, with MemGPT explicitly adopting an operating-system metaphor of paging across memory tiers [78–79]. A-MEM and MemoryOS moved toward agentic memory management: the memory system itself performs linking, updating, and routing rather than simply serving retrieval results [80–81]. Reflective Memory Management, LD-Agent, SeCom, retrieval-enhanced profile memory, TailorRPA, Bi-Mem, SGMem, and LightMem push further into hierarchical, reflective, graph-based, or computationally budgeted memory construction [82–89]. Across these papers, a convergent architecture is visible: short-term memory supports immediate coherence, mid-term memory stores reusable summaries or events, and long-term memory stores consolidated profiles, facts, or procedures. The design disagreement lies in

how to construct and maintain these layers—heuristic summarization, learned extraction, graph induction, reflective revision, or explicit lifecycle management. ¹²

A useful synthesis is to separate agent memory into four representational modes. **Episodic memory** stores events and trajectories; **semantic memory** stores facts, summaries, relations, or profiles; **procedural memory** stores skills, code, policies, and action templates; **reflective memory** stores critiques, lessons, or abstractions from multiple episodes. Many systems combine several modes. Voyager’s skill library is procedural [73]; Reflexion’s verbal reinforcement is reflective [71]; MemoryBank is largely episodic-semantic [78]; profile-memory systems are semantic [85–86]; graph- or hierarchy-based systems such as HippoRAG, A-MEM, Bi-Mem, and SGMem attempt to connect episodic fragments to higher-level semantic organization [67, 80, 87–88]. The field’s more recent surveys increasingly describe progress as a transition from storing trajectories to extracting reusable experience, which seems directionally correct: raw storage alone scales poorly with horizon length, but over-aggressive abstraction can erase rare yet crucial exceptions [16–17]. ¹³

Compression, selection, and memory control form the fourth family. LLMingua, LongLLMingua, and LLMingua-2 compress prompts by selecting more informative content while trying to preserve task fidelity [123–125]. Selective Context and Activation Beacon pursue similar goals through information-based filtering or activation-level compression [126–127], while gist-token approaches offer learned compact surrogates for lengthy prompts [128]. At a conceptual level, these methods implement a “what is worth keeping?” policy before or during memory writing. They are especially relevant to agents because long histories typically consist of mostly low-value routine tokens interspersed with a few decisive facts, constraints, or failures. Nevertheless, compression remains brittle: importance is query-dependent, future tasks are unknown, and the best summary for response generation may be the wrong summary for later planning or debugging. This is why several recent works evaluate long-context methods through shared-context and multi-turn settings rather than single-query recall alone [109, 123–128]. ¹⁴

Evaluation methodology has become a research area in its own right. ZeroSCROLLS and LongBench broadened early long-text evaluation beyond synthetic retrieval [98–99], while L-Eval, InfiniteBench, RULER, BABILong, NeedleBench, NoLiMa, HELMET, LongBench v2, AcademicEval, and SCBench progressively exposed different failure modes: position bias, lexical shortcutting, weak generation, shallow reasoning, unrealistic metric choices, and shared-context degradation [100–109, 129–132]. For agent memory specifically, LoCoMo, LongMemEval, LongMemEval-V2, LoCoMo-Plus, MemoryAgentBench, and MemoryCD move evaluation into multi-session interaction, cognitive memory, and lifelong personalization [90–95]. The broad lesson is that long-horizon competence is multidimensional. A system may retrieve explicit facts correctly yet fail to preserve latent constraints; it may perform well when the answer string is unique yet fail when semantic associations must be reconstructed; it may excel in single-turn benchmarks and collapse when memory must survive many future queries. Benchmark design now increasingly reflects these distinctions, and any serious claim about “memory” should specify which aspect is being measured. ¹⁵

Finally, **downstream agent environments** matter because memory quality is only meaningful insofar as it improves decision-making over extended horizons. AgentBench, WebArena, OSWorld, WorkArena, WorkArena++, Mind2Web, WebShop, ALFWorld, API-Bank, GAIA, AutoGen, Toolformer, and SWE-bench expose different requirements for remembering prior observations, preserving execution state, and reusing past solutions [110–122]. In software engineering and enterprise environments especially, the horizon is not just long in tokens; it is long in dependency structure. Solving a SWE-bench or WorkArena++ task may require combining multi-file state, unresolved subgoals, tool outputs, and environment-specific conventions. This is precisely the setting where the distinction between a large context window and a maintained agent memory becomes operational rather than definitional. ¹⁶

Challenges and Prospects

The most immediate challenge is **memory write quality**. Current systems are better at storage than at deciding what should be stored and in what representation. Raw transcript storage preserves fidelity but scales poorly. Aggressive summarization reduces load but risks flattening nuance, exceptions, uncertainty, and temporal ordering. Graph- and hierarchy-based systems improve structure, but they can crystallize early extraction errors and propagate hallucinated relations through consolidation. This makes write-time supervision, revision, and provenance tracing at least as important as retrieval-time ranking [78–89, 97]. A major open problem is learning memory writing policies that optimize for future usefulness rather than present salience. ¹⁷

A second challenge is **state alignment across memory layers**. Long-horizon agents now commonly maintain several coexisting states: the current prompt, the model’s KV cache, an external vector or graph store, and semi-structured summaries or profiles. These states can drift apart. For example, a memory retriever may return semantically relevant but temporally stale facts; a compressed KV cache may contain information optimized for the current query but hostile to later turns; a profile memory may overgeneralize from transient behavior. Recent shared-context and cost-performance studies reinforce that no single layer dominates across workloads, and localized memory maintenance often beats indiscriminate global reorganization [96–97, 109]. This suggests that future systems will need explicit synchronization and consistency protocols, not just better retrievers. ¹⁸

A third challenge is **credit assignment over delayed horizons**. Many memory papers evaluate whether a past fact can be recalled, yet the harder question is whether memory changes later action quality. An agent may remember that an interface contains a hidden menu, but the key competence is to exploit that memory at the right future moment. This suggests a shift from memory evaluation as question answering toward memory evaluation as policy improvement. LongMemEval-V2, MemoryAgentBench, MemoryCD, and agent-native systems studies are important steps because they frame memory as a contributor to later competence in dynamic environments, not merely as a retrieval service [92–97]. A promising direction is to treat memory operations—write, merge, delete, route, and revise—as controllable decisions with explicit downstream utility. ¹⁹

A fourth challenge is **robustness, privacy, and security**. Persistent agent memory changes the threat model of LLM systems. Sensitive facts can be stored longer than intended, malicious users can poison future interactions, and stale or incorrect memories can continue to influence decisions after the original context is gone. The literature is beginning to address these issues, but the empirical base remains much thinner than the performance literature. At a minimum, future memory systems need provenance, editability, temporal scoping, user-level isolation, and reliable abstention when retrieved memory is uncertain or conflicting [79–81, 89, 91–97]. For many real deployments, governance may become a first-order design constraint rather than an afterthought. ²⁰

A fifth challenge is **theoretical understanding**. The field still lacks a satisfying account of when long context should dominate explicit memory, when retrieval should dominate compression, and when procedures or reflections should be stored instead of facts. Empirically, the answer appears workload-dependent: architectural long-context methods are attractive when the immediate input is large and mostly relevant; external memory is attractive when persistent history is large but sparsely relevant; hierarchical or reflective memories are attractive when later decisions depend on abstractions rather than verbatim recall [41–50, 64–69, 90–109]. What is missing is a principled theory relating task structure, horizon length, uncertainty, and memory representation. Without such a theory, system design remains heavily benchmark-driven. ²¹

A final and especially promising prospect is the emergence of **agent-native memory systems**. Rather than bolting a retriever or a database onto an LLM, these systems treat memory as a managed resource with lifecycle operations: ingestion, abstraction, indexing, routing, revision, forgetting, and reuse. The strongest recent papers increasingly move in this direction, but the evidence remains incomplete. Several results are still reported on limited domains, and many papers do not separate gains from better retrieval, better summaries, more powerful base models, or prompt engineering. Accordingly, the central limitation of the present literature is not a shortage of ideas but a shortage of controlled causal evaluation. The next phase of the field will likely be defined less by ever-larger windows and more by reproducible, workload-aware memory systems whose behavior can be decomposed and audited [80–97]. ²²

Conclusion

Memory and long-context mechanisms are converging into a single research program aimed at giving LLM agents durable, selective, and computationally tractable state over long interaction horizons. Architectural methods enlarge or restructure working memory; external retrieval methods provide updateable persistence; agentic memory systems transform trajectories into reusable experience; compression and cache methods control cost; and new benchmarks increasingly reveal that these layers solve different problems rather than substituting for one another [13–17, 33–50, 51–69, 70–97, 98–132].

²³

The best current evidence supports a layered design principle for long-horizon agents. A strong agent should combine a reasonably capable long-context backbone, explicit persistent memory, structured consolidation, and workload-specific evaluation. Relying on only one of these components creates predictable failure modes: pure long-context systems incur cost and position bias; pure retrieval systems incur staleness and query mismatch; pure summarization systems risk abstraction loss; and pure benchmark optimization risks overfitting to superficial recall. The mature solution is therefore not “long context versus memory,” but **long context for bandwidth, memory for persistence, and abstraction for reuse**. The most important open question is how to make these layers cooperate under realistic latency, privacy, and reliability constraints in the wild [12, 41, 67–69, 90–97, 109–122]. ²⁴

References

- [1] Jeffrey L. Elman. Finding structure in time. *Cognitive Science*, 1990.
- [2] Sepp Hochreiter and Jürgen Schmidhuber. Long short-term memory. *Neural Computation*, 1997.
- [3] Alex Graves, Greg Wayne, and Ivo Danihelka. Neural Turing Machines. *arXiv:1410.5401*, 2014.
- [4] Jason Weston, Sumit Chopra, and Antoine Bordes. Memory Networks. *ICLR*, 2015. *arXiv:1410.3916*.
- [5] Sainbayar Sukhbaatar, Arthur Szlam, Jason Weston, and Rob Fergus. End-To-End Memory Networks. *NeurIPS*, 2015. *arXiv:1503.08895*.
- [6] Alex Graves, Greg Wayne, Malcolm Reynolds, et al. Hybrid computing using a neural network with dynamic external memory. *Nature*, 2016. *arXiv:1603.04467*.
- [7] Ashish Vaswani, Noam Shazeer, Niki Parmar, et al. Attention Is All You Need. *NeurIPS*, 2017.

- [8] Zihang Dai, Zhilin Yang, Yiming Yang, et al. Transformer-XL: Attentive Language Models beyond a Fixed-Length Context. ACL, 2019. arXiv:1901.02860.
- [9] Jack W. Rae, Anna Potapenko, Siddhant M. Jayakumar, and Timothy P. Lillicrap. Compressive Transformers for Long-Range Sequence Modelling. ICLR, 2020. arXiv:1911.05507.
- [10] Ofir Press, Noah A. Smith, and Mike Lewis. Train Short, Test Long: Attention with Linear Biases Enables Input Length Extrapolation. ICLR, 2022. arXiv:2108.12409.
- [11] Jianlin Su, Yu Lu, Shengfeng Pan, Ahmed Murtadha, and Yunfeng Liu. RoFormer: Enhanced Transformer with Rotary Position Embedding. Neurocomputing, 2024. arXiv:2104.09864.
- [12] Nelson F. Liu, Kevin Lin, John Hewitt, et al. Lost in the Middle: How Language Models Use Long Contexts. TACL, 2024. arXiv:2307.03172.
- [13] Jiaheng Liu, Dawei Zhu, Wenhao Huang, et al. A Comprehensive Survey on Long Context Language Modeling. arXiv:2503.17407, 2025.
- [14] Zhicheng Zhang, Xinyue He, et al. A Survey on the Memory Mechanism of LLM-based Agents. arXiv:2404.13501, 2024.
- [15] Tengen Du, Bowen Song, et al. Memory for Autonomous LLM Agents. arXiv:2602.05929, 2026.
- [16] Yuanzhe Hu, Yu Wang, and Julian McAuley. Memory in the Age of AI Agents: Current Approaches and Open Challenges. arXiv:2504.02441, 2025.
- [17] Jinghao Luo, Yuchen Tian, Chuxue Cao, et al. From Storage to Experience: A Survey on the Evolution of LLM Agent Memory Mechanisms. arXiv:2605.06716, 2026.
- [18] Rewon Child, Scott Gray, Alec Radford, and Ilya Sutskever. Generating Long Sequences with Sparse Transformers. arXiv:1904.10509, 2019.
- [19] Nikita Kitaev, Łukasz Kaiser, and Anselm Levskaya. Reformer: The Efficient Transformer. ICLR, 2020. arXiv:2001.04451.
- [20] Iz Beltagy, Matthew E. Peters, and Arman Cohan. Longformer: The Long-Document Transformer. arXiv:2004.05150, 2020.
- [21] Manzil Zaheer, Guru Guruganesh, Avinava Dubey, et al. Big Bird: Transformers for Longer Sequences. NeurIPS, 2020. arXiv:2007.14062.
- [22] Sinong Wang, Belinda Z. Li, Madian Khabsa, Han Fang, and Hao Ma. Linformer: Self-Attention with Linear Complexity. arXiv:2006.04768, 2020.
- [23] Angelos Katharopoulos, Apoorv Vyas, Nikolaos Pappas, and François Fleuret. Transformers are RNNs: Fast Autoregressive Transformers with Linear Attention. ICML, 2020. arXiv:2006.16236.
- [24] Krzysztof Choromanski, Valerii Likhoshesterov, David Dohan, et al. Rethinking Attention with Performers. ICLR, 2021. arXiv:2009.14794.

- [25] Yi Tay, Mostafa Dehghani, Samira Abnar, et al. Long Range Arena: A Benchmark for Efficient Transformers. ICLR, 2021. arXiv:2011.04006.
- [26] Albert Gu, Karan Goel, and Christopher Ré. Efficiently Modeling Long Sequences with Structured State Spaces. ICLR, 2022. arXiv:2111.00396.
- [27] Erhan Poli, Aran C. Cooper, et al. Hyena Hierarchy: Towards Larger Convolutional Language Models. ICML, 2023. arXiv:2302.10866.
- [28] Yutao Sun, Li Dong, et al. Retentive Network: A Successor to Transformer for Large Language Models. arXiv:2307.08621, 2023.
- [29] Albert Gu and Tri Dao. Mamba: Linear-Time Sequence Modeling with Selective State Spaces. COLM, 2024. arXiv:2312.00752.
- [30] Oren Lieber, Or Sharir, Barak Lenz, and Yoav Shoham. Jamba: A Hybrid Transformer-Mamba Language Model. arXiv:2403.19887, 2024.
- [31] Tri Dao, Daniel Y. Fu, Stefano Ermon, Atri Rudra, and Christopher Ré. FlashAttention: Fast and Memory-Efficient Exact Attention with IO-Awareness. NeurIPS, 2022. arXiv:2205.14135.
- [32] Tri Dao. FlashAttention-2: Faster Attention with Better Parallelism and Work Partitioning. ICLR, 2024. arXiv:2307.08691.
- [33] Shouyuan Chen, Sherwin Chen, et al. Extending Context Window of Large Language Models via Positional Interpolation. arXiv:2306.15595, 2023.
- [34] Bowen Peng, Jeffrey Quesnelle, Honglu Fan, and Enrico Shippole. YaRN: Efficient Context Window Extension of Large Language Models. ICLR, 2024. arXiv:2309.00071.
- [35] Yukang Chen, Shengju Qian, Haotian Tang, et al. LongLoRA: Efficient Fine-tuning of Long-Context Large Language Models. ICLR, 2024. arXiv:2309.12307.
- [36] Jianxin Yang. LongQLoRA: Efficient and Effective Method to Extend Context Length of Large Language Models. arXiv:2311.04879, 2023.
- [37] Hongyu Jin, Xizhang Han, et al. Self-Extend: Self-Extend LLM Context Window Without Tuning. arXiv: 2401.01325, 2024.
- [38] Yiran Ding, Siyue Gao, et al. LongRoPE: Extending LLM Context Window Beyond 2 Million Tokens. arXiv:2402.13753, 2024.
- [39] Na Shang, Yiran Ding, et al. LongRoPE2: Near-Lossless LLM Context Window Scaling. arXiv: 2502.20082, 2025.
- [40] Zhiyuan Hu, Yuliang Liu, et al. LongRecipe: Recipe for Efficient Long Context Generalization in Large Language Models. arXiv:2409.00509, 2024.

- [41] Tianyu Gao, Howard Yen, and Danqi Chen. How to Train Long-Context Language Models (Effectively). ACL, 2025. arXiv:2410.02660.
- [42] Howard Yen, Tianyu Gao, and Danqi Chen. Long-Context Language Modeling with Parallel Context Encoding. ACL, 2024. arXiv:2402.16617.
- [43] Tsendsuren Munkhdalai, Manaal Faruqui, and Siddharth Gopal. Leave No Context Behind: Efficient Infinite Context Transformers with Infini-attention. arXiv:2404.07143, 2024.
- [44] Chaojun Xiao, Pengle Zhang, Xu Han, et al. InfLLM: Training-Free Long-Context Extrapolation for LLMs with an Efficient Context Memory. NeurIPS, 2024. arXiv:2402.04617.
- [45] Amirkeivan Mohtashami and Martin Jaggi. Landmark Attention: Random-Access Infinite Context Length for Transformers. arXiv:2305.16300, 2023.
- [46] Arka Pal, Deep Karkhanis, Manley Roberts, et al. Giraffe: Adventures in Expanding Context Lengths in LLMs. arXiv:2308.10882, 2023.
- [47] Guangxuan Xiao, Yuandong Tian, Beidi Chen, Song Han, and Mike Lewis. Efficient Streaming Language Models with Attention Sinks. arXiv:2309.17453, 2023.
- [48] Zhenyu Zhang, Ying Sheng, Tianyi Zhou, et al. H2O: Heavy-Hitter Oracle for Efficient Generative Inference of Large Language Models. NeurIPS, 2023.
- [49] Yuhong Li, Yingbing Huang, Bowen Yang, et al. SnapKV: LLM Knows What You are Looking for Before Generation. arXiv:2404.14469, 2024.
- [50] Huiqiang Jiang, Yucheng Li, Chengruidong Zhang, et al. MInference 1.0: Accelerating Pre-filling for Long-Context LLMs via Dynamic Sparse Attention. arXiv:2407.02490, 2024.
- [51] Urvashi Khandelwal, Omer Levy, Dan Jurafsky, Luke Zettlemoyer, and Mike Lewis. Generalization through Memorization: Nearest Neighbor Language Models. ICLR, 2020. arXiv:1911.00172.
- [52] Kelvin Guu, Kenton Lee, Zora Tung, Panupong Pasupat, and Ming-Wei Chang. REALM: Retrieval-Augmented Language Model Pre-Training. ICML, 2020. arXiv:2002.08909.
- [53] Patrick Lewis, Ethan Perez, Aleksandra Piktus, et al. Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks. NeurIPS, 2020. arXiv:2005.11401.
- [54] Gautier Izacard and Edouard Grave. Leveraging Passage Retrieval with Generative Models for Open Domain Question Answering. EACL, 2021. arXiv:2007.01282.
- [55] Gautier Izacard, Fabio Petroni, et al. Atlas: Few-shot Learning with Retrieval Augmented Language Models. JMLR, 2023. arXiv:2208.03299.
- [56] Sebastian Borgeaud, Arthur Mensch, Jordan Hoffmann, et al. Improving Language Models by Retrieving from Trillions of Tokens. ICML, 2022. arXiv:2112.04426.

- [57] Yuhuai Wu, Markus N. Rabe, DeLesley Hutchins, and Christian Szegedy. Memorizing Transformers. ICLR, 2022. arXiv:2203.08913.
- [58] Qingyang Wu, Zhenzhong Lan, Kun Qian, et al. Memformer: A Memory-Augmented Transformer for Sequence Modeling. Findings of AACL, 2022. arXiv:2010.06891.
- [59] Weizhi Wang, Li Dong, Hao Cheng, et al. Augmenting Language Models with Long-Term Memory. NeurIPS, 2023. arXiv:2306.07174.
- [60] Yu Wang, Yifan Gao, Xiusi Chen, et al. MEMORYLLM: Towards Self-Updatable Large Language Models. arXiv:2402.04624, 2024.
- [61] Yu Wang, Yifan Gao, et al. Extending MemoryLLM with Scalable Long-Term Memory. arXiv:2502.00592, 2025.
- [62] Weijie Liu, Zecheng Tang, Juntao Li, Kehai Chen, and Min Zhang. MemLong: Memory-Augmented Retrieval for Long Text Modeling. arXiv:2408.16967, 2024.
- [63] Phoebe Klett and Thomas Ahle. Extended Mind Transformers. arXiv:2406.02332, 2024.
- [64] Weijia Shi, Sewon Min, Michihiro Yasunaga, et al. REPLUG: Retrieval-Augmented Black-Box Language Models. arXiv:2301.12652, 2023.
- [65] Akari Asai, Zeqiu Wu, Yizhong Wang, Avirup Sil, and Hannaneh Hajishirzi. Self-RAG: Learning to Retrieve, Generate, and Critique through Self-Reflection. ICLR, 2024. arXiv:2310.11511.
- [66] Parth Sarthi, Salman Abdullah, Aditi Tuli, et al. RAPTOR: Recursive Abstractive Processing for Tree-Organized Retrieval. arXiv:2401.18059, 2024.
- [67] Bryan J. Gutiérrez, Yu Su, et al. HippoRAG: Neurobiologically Inspired Long-Term Memory for Large Language Models. arXiv:2405.14831, 2024.
- [68] Haotian Qian, et al. MemoRAG: Boosting Long Context Processing with Global Memory-Augmented Retrieval. arXiv:2409.05591, 2024.
- [69] Zirui Guo, Lianghao Xia, Yanhua Yu, Tu Ao, and Chao Huang. LightRAG: Simple and Fast Retrieval-Augmented Generation. arXiv:2410.05779, 2024.
- [70] Joon Sung Park, Joseph C. O’ Brien, Carrie J. Cai, et al. Generative Agents: Interactive Simulacra of Human Behavior. UIST, 2023. arXiv:2304.03442.
- [71] Noah Shinn, Federico Cassano, Edward Berman, Ashwin Gopinath, Karthik Narasimhan, and Shunyu Yao. Reflexion: Language Agents with Verbal Reinforcement Learning. NeurIPS, 2023. arXiv:2303.11366.
- [72] Andrew Zhao, et al. ExpeL: LLM Agents Are Experiential Learners. arXiv:2308.10144, 2023.
- [73] Guanzhi Wang, Yuqi Xie, Yunfan Jiang, et al. Voyager: An Open-Ended Embodied Agent with Large Language Models. arXiv:2305.16291, 2023.

- [74] Shunyu Yao, Jeffrey Zhao, Dian Yu, et al. ReAct: Synergizing Reasoning and Acting in Language Models. ICLR, 2023. arXiv:2210.03629.
- [75] Shunyu Yao, Dian Yu, Jeffrey Zhao, et al. Tree of Thoughts: Deliberate Problem Solving with Large Language Models. NeurIPS, 2023. arXiv:2305.10601.
- [76] Shibo Hao, Yi Gu, Haodi Ma, et al. Reasoning with Language Model is Planning with World Model. EMNLP, 2023. arXiv:2305.14992.
- [77] Andy Zhou, Kai Yan, Michal Shlapentokh-Rothman, Haohan Wang, and Yu-Xiong Wang. Language Agent Tree Search Unifies Reasoning, Acting, and Planning in Language Models. ICML, 2024. arXiv:2310.04406.
- [78] Wanjun Zhong, Lianghong Guo, Qiqi Gao, He Ye, and Yanlin Wang. MemoryBank: Enhancing Large Language Models with Long-Term Memory. AAAI, 2024. arXiv:2305.10250.
- [79] Charles Packer, Sarah Wooders, Kevin Lin, Vivian Fang, Shishir G. Patil, Ion Stoica, and Joseph E. Gonzalez. MemGPT: Towards LLMs as Operating Systems. arXiv:2310.08560, 2023.
- [80] Wujiang Xu, Zujie Liang, Kai Mei, et al. A-MEM: Agentic Memory for LLM Agents. arXiv:2502.12110, 2025.
- [81] Jiazheng Kang, Mingming Ji, Zhe Zhao, and Ting Bai. Memory OS of AI Agent. arXiv:2506.06326, 2025.
- [82] Zhen Tan, Jun Yan, I-Hung Hsu, et al. In Prospect and Retrospect: Reflective Memory Management for Long-term Personalized Dialogue Agents. ACL, 2025. arXiv:2503.08026.
- [83] Hongxin Li, et al. Hello Again! LLM-powered Personalized Agent for Long-Term Dialogue. NAACL, 2025.
- [84] Zhuoshi Pan, Qianhui Wu, Huiqiang Jiang, et al. SeCom: On Memory Construction and Retrieval for Personalized Conversational Agents. ICLR, 2025.
- [85] Zixun Wang, et al. Crafting Personalized Agents through Retrieval-Enhanced Profile Memory. EMNLP, 2024.
- [86] Zhenpeng Gao, Xiaofen Xin, and Xiangmin Xu. TailorRPA: A Retrieval-Based Framework for Eliciting Personalized and Coherent Role-Playing Agents in General Domain. Findings of EMNLP, 2025.
- [87] Wenyu Mao, Haosong Tan, Shuchang Liu, et al. Bi-Mem: Bidirectional Construction of Hierarchical Memory for Personalized LLMs via Inductive-Reflective Agents. arXiv:2601.06490, 2026.
- [88] Yimin Wu, Yijie Zhang, Shuxin Liang, and Yeting Liu. SGMem: Sentence Graph Memory for Long-Term Conversational Agents. arXiv:2509.21212, 2025.
- [89] Jiaquan Fang, et al. Lightweight LLM Agent Memory with Small Language Models. arXiv:2604.07798, 2026.

- [90] Adyasha Maharana, Dongwon Lee, et al. Evaluating Very Long-Term Conversational Memory of LLM Agents. arXiv:2402.17753, 2024.
- [91] Di Wu, Hongwei Wang, Wenhao Yu, Yuwei Zhang, Kai-Wei Chang, and Dong Yu. LongMemEval: Benchmarking Chat Assistants on Long-Term Interactive Memory. arXiv:2410.10813, 2024.
- [92] Di Wu, Hongwei Wang, et al. LongMemEval-V2: Evaluating Long-Term Agent Memory. arXiv:2605.12493, 2026.
- [93] Yifei Li, Weidong Guo, Lingling Zhang, et al. LoCoMo-Plus: Beyond-Factual Cognitive Memory Evaluation Framework for LLM Agents. arXiv:2602.10715, 2026.
- [94] Yuanzhe Hu, Yu Wang, and Julian McAuley. Evaluating Memory in LLM Agents via Incremental Multi-Turn Interactions. arXiv:2507.05257, 2025.
- [95] Weizhi Zhang, Xiaokai Wei, Wei-Chieh Huang, et al. MemoryCD: Benchmarking Long-Context User Memory of LLM Agents for Lifelong Cross-Domain Personalization. arXiv:2603.25973, 2026.
- [96] Natchanon Pollertlam and Witchayut Kornsuwannawit. Beyond the Context Window: A Cost-Performance Analysis of Fact-Based Memory vs. Long-Context LLMs for Persistent Agents. arXiv:2603.04814, 2026.
- [97] Wei Zhou, Xuanhe Zhou, Shaokun Han, et al. Are We Ready For An Agent-Native Memory System? arXiv:2606.24775, 2026.
- [98] Uri Shaham, Maor Ivgi, Avia Efrat, Jonathan Berant, and Omer Levy. ZeroSCROLLS: A Zero-Shot Benchmark for Long Text Understanding. Findings of EMNLP, 2023. arXiv:2305.14196.
- [99] Yushi Bai, Xin Lv, et al. LongBench: A Bilingual, Multitask Benchmark for Long Context Understanding. arXiv:2308.14508, 2023.
- [100] Chenxin An, Shansan Gong, Ming Zhong, et al. L-Eval: Instituting Standardized Evaluation for Long Context Language Models. ACL, 2024. arXiv:2307.11088.
- [101] Xinrong Zhang, Yingfa Chen, Shengding Hu, et al. InfiniteBench: Extending Long Context Evaluation Beyond 100K Tokens. ACL, 2024. arXiv:2402.13718.
- [102] Cheng-Ping Hsieh, Simeng Sun, Samuel Krizan, et al. RULER: What’s the Real Context Size of Your Long-Context Language Models? arXiv:2404.06654, 2024.
- [103] Yuri Kuratov, Aydar Bulatov, Petr Anokhin, et al. BABILong: Testing the Limits of LLMs with Long Context Reasoning-in-a-Haystack. arXiv:2406.10149, 2024.
- [104] Mo Li, Songyang Zhang, Yunxin Liu, and Kai Chen. NeedleBench: Can LLMs Do Retrieval and Reasoning in 1 Million Context Window? arXiv:2407.11963, 2024.
- [105] Ali Modarressi, Hanieh Deilamsalehy, Franck Dernoncourt, et al. NoLiMa: Long-Context Evaluation Beyond Literal Matching. arXiv:2502.05167, 2025.

- [106] Howard Yen, Tianyu Gao, Minmin Hou, et al. HELMET: How to Evaluate Long-Context Language Models Effectively and Thoroughly. ICLR, 2025. arXiv:2410.02694.
- [107] Yushi Bai, Shangqing Tu, Jiajie Zhang, et al. LongBench v2: Towards Deeper Understanding and Reasoning on Realistic Long-Context Multitasks. arXiv:2412.15204, 2024.
- [108] Haozhen Zhang, Tao Feng, Pengrui Han, and Jiaxuan You. AcademicEval: Live Long-Context LLM Benchmark. arXiv:2510.17725, 2025.
- [109] Yucheng Li, Huiqiang Jiang, Qianhui Wu, et al. SCBench: A KV Cache-Centric Analysis of Long-Context Methods. arXiv:2412.10319, 2024.
- [110] Carlos E. Jimenez, John Yang, Alexander Wettig, et al. SWE-bench: Can Language Models Resolve Real-World GitHub Issues? ICLR, 2024. arXiv:2310.06770.
- [111] Xiao Liu, Hao Yu, Hanchen Zhang, et al. AgentBench: Evaluating LLMs as Agents. arXiv:2308.03688, 2023.
- [112] Shuyan Zhou, Frank F. Xu, Hao Zhu, et al. WebArena: A Realistic Web Environment for Building Autonomous Agents. ICLR, 2024. arXiv:2307.13854.
- [113] Tianbao Xie, et al. OSWorld: Benchmarking Multimodal Agents for Open-Ended Tasks in Real Computer Environments. arXiv:2404.07972, 2024.
- [114] Alexandre Drouin, et al. WorkArena: How Capable Are Web Agents at Solving Common Knowledge Work Tasks? arXiv:2403.07718, 2024.
- [115] Laurent Boisvert, et al. WorkArena++: Towards Compositional Planning and Reasoning for Web Agents. arXiv:2407.05291, 2024.
- [116] Xiang Deng, Yu Gu, Boyuan Zheng, et al. Mind2Web: Towards a Generalist Agent for the Web. NeurIPS, 2023. arXiv:2306.06070.
- [117] Shunyu Yao, Howard Chen, John Yang, and Karthik Narasimhan. WebShop: Towards Scalable Real-World Web Interaction with Grounded Language Agents. NeurIPS, 2022. arXiv:2207.01206.
- [118] Mohit Shridhar, Xavier Yuan, Marc Côté, et al. ALFWorld: Aligning Text and Embodied Environments for Interactive Learning. ICLR, 2021. arXiv:2010.03768.
- [119] Minghao Li, Yingxiu Zhao, Bowen Yu, et al. API-Bank: A Comprehensive Benchmark for Tool-Augmented LLMs. EMNLP, 2023. arXiv:2304.08244.
- [120] Grégoire Mialon, et al. GAIA: A Benchmark for General AI Assistants. arXiv:2311.12983, 2023.
- [121] Qingyun Wu, Gagan Bansal, Jieyu Zhang, et al. AutoGen: Enabling Next-Gen LLM Applications via Multi-Agent Conversation. arXiv:2308.08155, 2023.
- [122] Timo Schick, Jane Dwivedi-Yu, Roberto Dessì, et al. Toolformer: Language Models Can Teach Themselves to Use Tools. NeurIPS, 2023. arXiv:2302.04761.

- [123] Huiqiang Jiang, Qianhui Wu, Chin-Yew Lin, Yuqing Yang, and Lili Qiu. LLMingua: Compressing Prompts for Accelerated Inference of Large Language Models. arXiv:2310.05736, 2023.
- [124] Huiqiang Jiang, Qianhui Wu, Xufang Luo, et al. LongLLMingua: Accelerating and Enhancing LLMs in Long Context Scenarios via Prompt Compression. arXiv:2310.06839, 2023.
- [125] Zhuoshi Pan, Qianhui Wu, Huiqiang Jiang, et al. LLMingua-2: Data Distillation for Efficient and Faithful Task-Agnostic Prompt Compression. Findings of ACL, 2024. arXiv:2403.12968.
- [126] Yucheng Li, Bo Dong, Chenghua Lin, and Frank Guerin. Compressing Context to Enhance Inference Efficiency of Large Language Models. arXiv:2310.06201, 2023.
- [127] Peitian Zhang, Zheng Liu, Shitao Xiao, et al. Long Context Compression with Activation Beacon. ICLR, 2025. arXiv:2401.03462.
- [128] Jesse Mu, Xiang Lisa Li, and Noah Goodman. Learning to Compress Prompts with Gist Tokens. arXiv:2304.08467, 2023.
- [129] Xiang Liu, Peijie Dong, Xuming Hu, and Xiaowen Chu. LongGenBench: Long-context Generation Benchmark. arXiv:2410.04199, 2024.
- [130] Marzena Karpinska, Katherine Thai, Kyle Lo, Tanya Goyal, and Mohit Iyyer. One Thousand and One Pairs: A “novel” Challenge for Long-Context Language Models. EMNLP, 2024. arXiv:2406.16264.
- [131] Simeng Sun, Kalpesh Krishna, Andrew Mattarella-Micke, and Mohit Iyyer. Do Long-Range Language Models Actually Use Long-Range Context? EMNLP, 2021.
- [132] Chonghua Wang, Haodong Duan, Songyang Zhang, Dahua Lin, and Kai Chen. Ada-LEval: Evaluating Long-Context LLMs with Length-Adaptable Benchmarks. NAACL, 2024. arXiv:2404.06480.

1 6 23 A Comprehensive Survey on Long Context Language Modeling

https://arxiv.org/abs/2503.17407?utm_source=chatgpt.com

2 7 <https://arxiv.org/abs/2404.07143>

<https://arxiv.org/abs/2404.07143>

3 5 24 <https://arxiv.org/abs/2307.03172>

<https://arxiv.org/abs/2307.03172>

4 11 <https://arxiv.org/abs/2304.03442>

<https://arxiv.org/abs/2304.03442>

8 <https://arxiv.org/abs/2404.14469>

<https://arxiv.org/abs/2404.14469>

9 <https://arxiv.org/abs/1911.00172>

<https://arxiv.org/abs/1911.00172>

10 <https://arxiv.org/abs/2405.14831>

<https://arxiv.org/abs/2405.14831>

- 12 <https://arxiv.org/abs/2305.10250>
<https://arxiv.org/abs/2305.10250>
- 13 <https://arxiv.org/abs/2605.06716>
<https://arxiv.org/abs/2605.06716>
- 14 <https://arxiv.org/abs/2310.05736>
<https://arxiv.org/abs/2310.05736>
- 15 LongBench: A Bilingual, Multitask Benchmark for Long Context Understanding
https://arxiv.org/abs/2308.14508?utm_source=chatgpt.com
- 16 <https://arxiv.org/abs/2308.03688>
<https://arxiv.org/abs/2308.03688>
- 17 <https://arxiv.org/abs/2503.08026>
<https://arxiv.org/abs/2503.08026>
- 18 <https://arxiv.org/abs/2603.04814>
<https://arxiv.org/abs/2603.04814>
- 19 <https://arxiv.org/abs/2605.12493>
<https://arxiv.org/abs/2605.12493>
- 20 <https://arxiv.org/abs/2310.08560>
<https://arxiv.org/abs/2310.08560>
- 21 <https://arxiv.org/html/2410.02660v2>
<https://arxiv.org/html/2410.02660v2>
- 22 <https://arxiv.org/abs/2502.12110>
<https://arxiv.org/abs/2502.12110>